

"Why is it that light waves with higher energies tend to be bluer?"

LEC15 - Optical Telescopes

Learning Goals

- Define properties of telescopes that affect what they are able to see
- Describe the types of telescopes and their pros, cons, features, and uses

Refraction & Reflection:

- Curved glass bends and focuses light to form an image
- Cameras are similar to optical and near-IR telescopes

Main Properties of Telescopes:

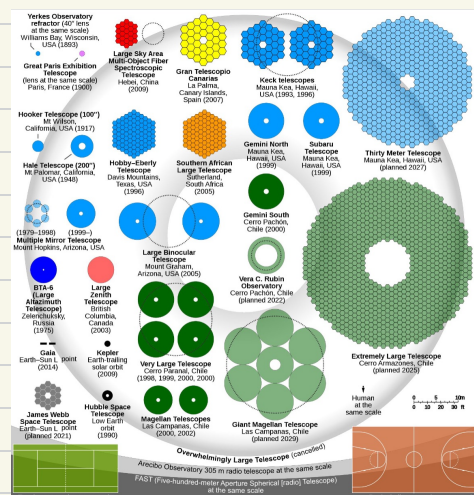
1. Collecting area

→ How many photons can the telescope collect per second

2. Resolution

→ How crisp of an image can it produce

- Both depend on size of telescope
- For telescopes, size = diameter of primary optic



Collecting Area: The area on a telescope where photons enter

Resolution: The ability for a telescope to separate 2 point sources into separate images. In an ideal environment (above atmosphere etc.) the resolution is bound by the diffraction limit

Light Collecting Area and Brightness:

- Collecting area determines how many photons can be collected per second
- Larger telescopes have larger collecting areas
- Basic purpose of a telescope is to collect many photons
- Larger collecting area means you can see fainter objects

Modern Research Telescopes:

- Largest mirrors are constructed from smaller segments
- Radio telescopes do not require large precise mirrors due to large wavelength of radio waves (easily reflected)

Example:

Telescope A has a diameter of 1m, Telescope B has a diameter of 10m, how much more photons does Telescope B collect per second than Telescope A?

$$A = \pi r^2 \Rightarrow A = \pi (10r)^2 \Rightarrow A = \pi 100r^2 \Rightarrow 100 \times$$

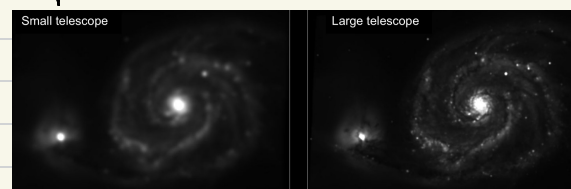
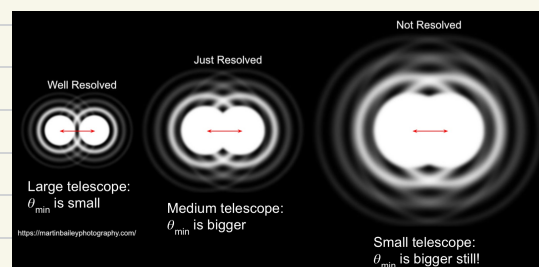
* It is impossible to enhance an image past its $\theta_{\min}$, since this is the limit

Resolution:

- What **angular separation** can we distinguish that there are two stars, not just one
- Because of diffraction, the best angular resolution for a telescope is related to the wavelength of light and size (diameter) of the telescope

$$\theta_{\min} = (2.5 \times 10^5 \text{ arcsec}) \cdot \frac{\lambda}{D}$$

- As telescope size increases, $\theta_{\min}$ decreases
- Different wavelengths show us different features



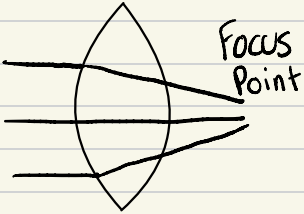
- Telescopes work by either reflecting or refracting light from a source (starlight) into your eyes
- Telescopes have 2 main properties. First is the size/diameter of the collecting area. Second is the resolution of the image produced.
- The minimum angular separation of 2 distinct points is given by $\theta_{\min} = (2.5 \times 10^5 \text{ arcsecond}) \cdot \frac{\lambda}{D}$

LEC15 - Optical Telescopes

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Lens

Focus Point



Magnification:

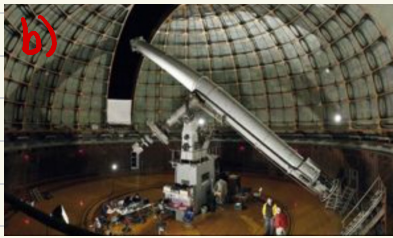
- We can use additional lenses to control the magnification and field

Refracting Telescopes:

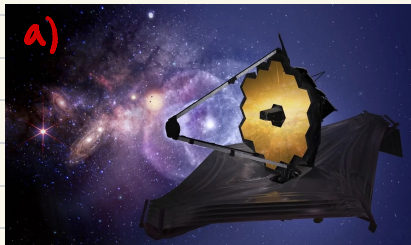
- Use lenses to focus light
- Lenses were easier to make, first telescopes were refracting telescopes
- Refractors are very long, since light follows a fairly straight path through them
- Lenses larger than ~1m are very impractical (Super heavy, sags under its own weight)

Reflecting Telescopes:

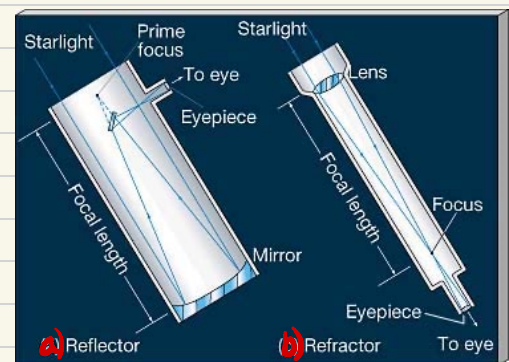
- Use mirrors to focus light
- Mirrors are made of glass with a thin coating of metal (silver, aluminum etc.)
- Large modern telescopes are reflectors
- Large mirrors can also sag under their own weight, or due to temperature/wind
 - Solution is to build smaller mirror segments, and use computers, sensors, and motors to control their shape
- Our largest telescopes are segmented



← Refracting Telescope



← Reflecting Telescope



- Refracting telescopes bend light waves into your eye, but are generally not used anymore since larger refracting telescopes are very difficult to produce
- Reflecting telescopes reflect light into your eye using mirrors. Modern large ones are made of smaller mirrors that have a computer align them so they act as a single large mirror